Experiment No:7 Continuous-Time Fourier Transform and Spectrum

Objective:

i. To determine the magnitude and phase spectra of continuous-time signals and analyze their significant spectral components.

ii. To analyze signal bandwidth and spectral components and relate the frequency-domain characteristics to communication system applications.

(i)To determine the magnitude and phase spectra of continuous-time signals and analyze their significant spectral components.

Procedure:

• Define t and the sampling interval.

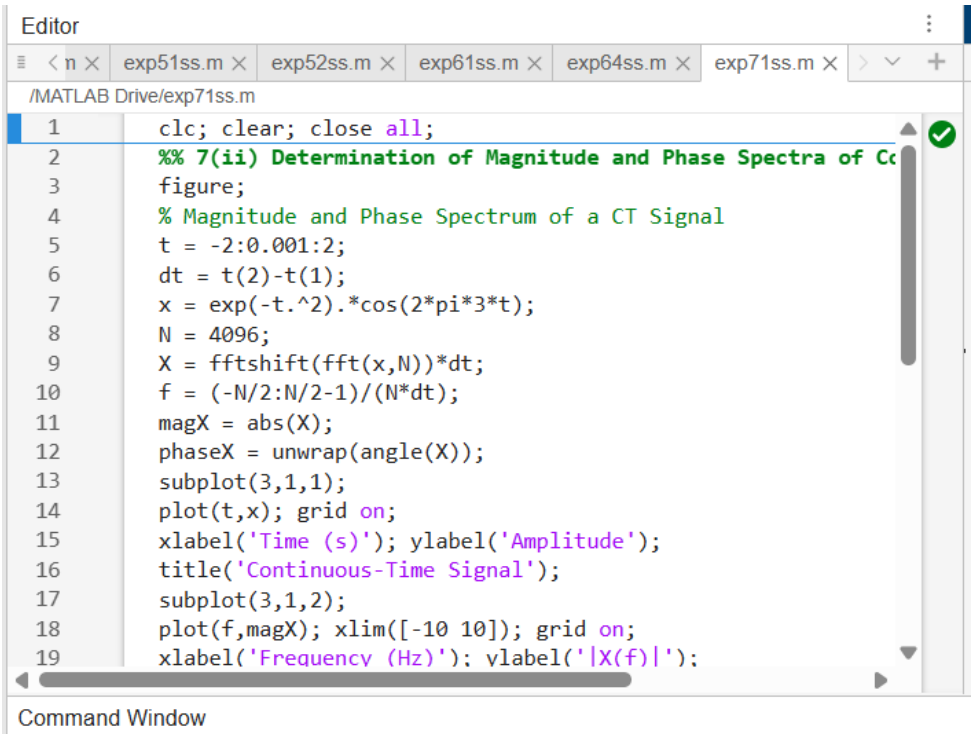
• Generate x(t)=e^(−t2)cos(2π3t).

• Compute the FFT-based transform with N = 4096.

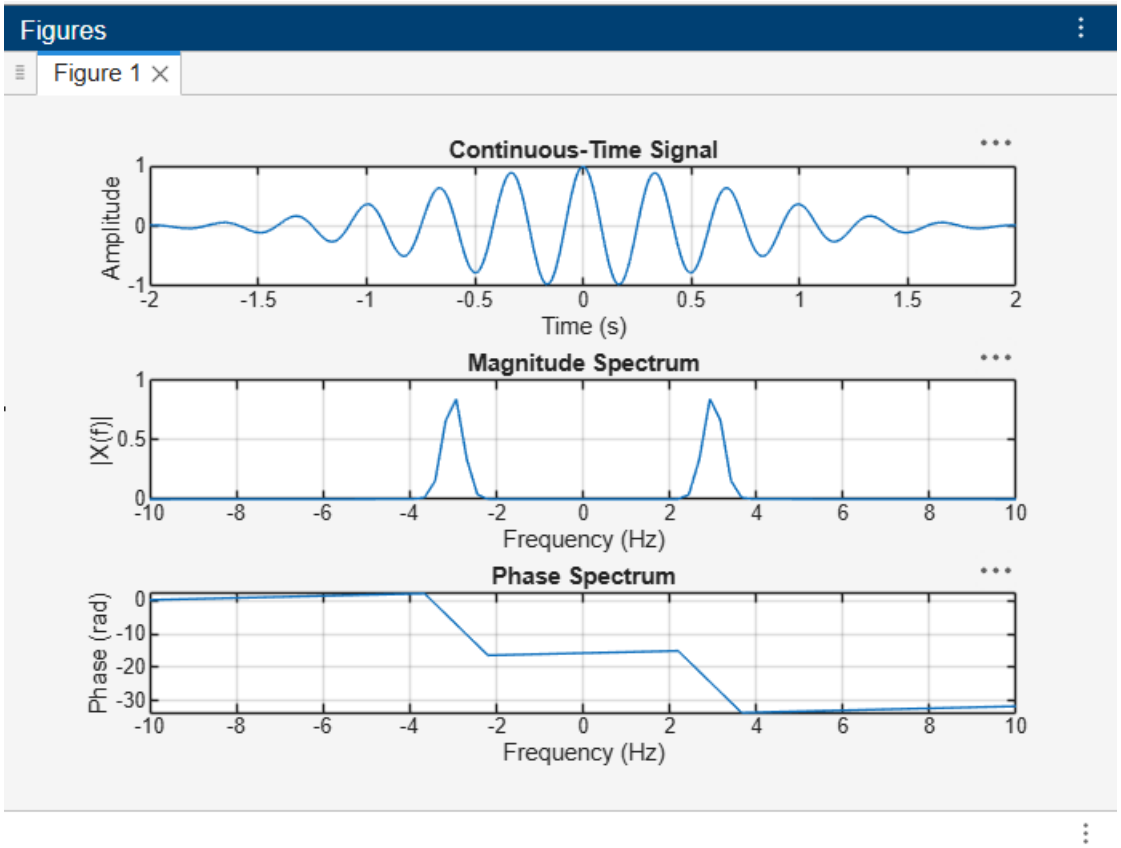
• Plot the signal, magnitude and unwrapped phase.

• Identify the principal spectral peaks.

Matlab Coding:



Output:



(iv) Analysis of Signal Bandwidth and Spectral Components for Communication Applications

Experimental Procedure:

• Set fs = 2000 Hz and generate the composite signal.

• Compute the FFT and shift the spectrum.

• Plot the magnitude spectrum from −400 to 400 Hz.

• Apply the 5% magnitude threshold.

• Determine the approximate one-sided occupied bandwidth.

MATLAB Coding:

Editor

exp52ss.m | exp61ss.m | exp64ss.m | exp71ss.m | exp72ss.m

/MATLAB Drive/exp72ss.m

```
clc;
clear;
close all;

% 7(iv): Signal Bandwidth and Spectral Components

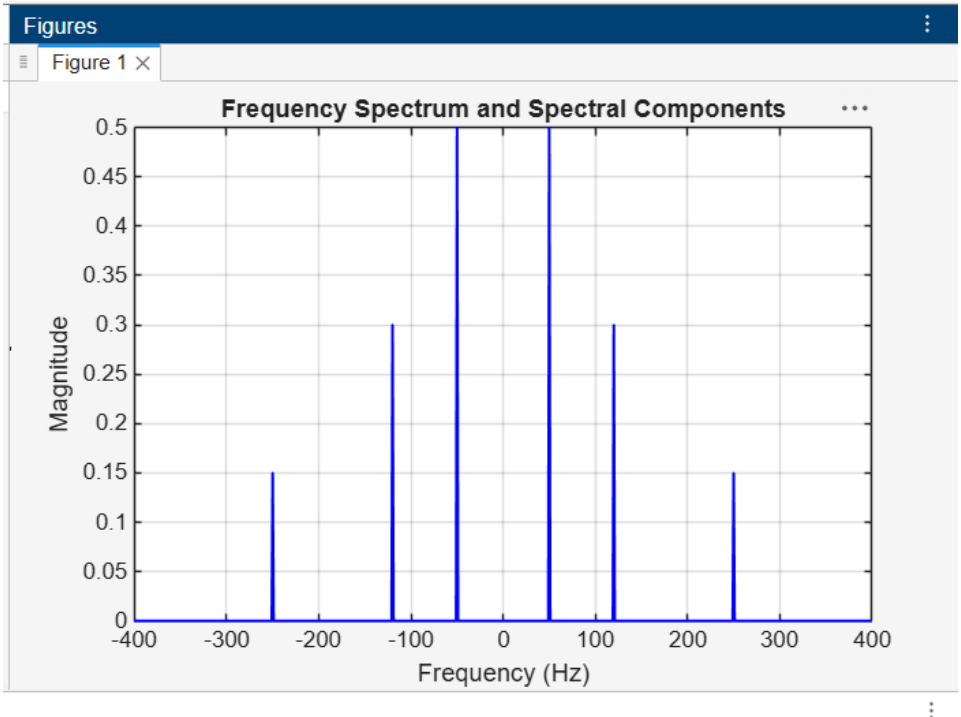
fs = 2000;
t = 0:1/fs:1-1/fs;

% Generate baseband signal
x = cos(2*pi*50*t) + 0.6*cos(2*pi*120*t) ...
    + 0.3*cos(2*pi*250*t);

% FFT analysis
N = length(x);
X = fftshift(fft(x))/N;
f = (-N/2:N/2-1)*(fs/N);
M = abs(X);
```

Command Window

Output:



Conclusion:

The continuous-time Fourier transform of representative signals was successfully obtained using FFT-based approximation, and their magnitude and phase spectra were analyzed to identify significant frequency components.The time-shift property was verified by comparing the computed and theoretical Fourier transforms, confirming that time shifting preserves magnitude while introducing a frequency-dependent phase shift.The spectral components and occupied bandwidth of a composite communication signal were also determined, demonstrating the importance of frequency-domain analysis in communication-system design.